

# Do Duc Anh Nguyen

✉ anhndd1510@gmail.com  
☎ (+33) 7 43 01 18 94  
35510 Cesson-Sévigné, France  
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## Education

2020–2022 **Master of Security Information**, University of Limoges, France

GPA M1: 14.877/20 - Rank 4/20

GPA M2: 14.836/20 - Rank 5/26

2015–2019 **Bachelor of Engineering, Computer Science**, University of Technology (HCMUT), Vietnam

GPA: 7.94/10 - HONORS PROGRAM

- **Traffic Light Control System (Thesis):** Using Deep Reinforcement Learning to implement a model to control the traffic light aiming to reduce the waiting time of vehicles. Built a simulation system, analyzed different models and training strategies.

Score: 9.6/10

## Working Experience

2023–Present **Doctoral Candidate in Information Security**, IMT Atlantique, Rennes, France

- **Research on Security Reactions to Fast Attack Events**

1. Investigating microservice-based architectures in decentralized environments to support rapid deployment of reaction mechanisms
2. Designing signaling methodologies to enable efficient information exchange between network nodes
3. Addressing limitations of conventional reaction systems, which often rely on centralized architectures and struggle to respond effectively to fast-propagating attacks
4. Studying various malware types and selecting representative use cases based on fast propagation characteristics

- **Lab teaching in Network Security: 21 hours**

Prepare and evaluate labs for ~ 20 students per lab

2022 **Master Intern**, IMT Atlantique, Rennes, France

- **Research on Intent-Based Networking (IBN)** Select the use case of existing IBN framework, Interface to Network Security Function (I2NSF), to evaluate our policy conflict detection mechanism performance

2018–2020 **Backend Developer**, Zalo Group, VNG Corporation, HCMC, Vietnam

- **Major servers building**

Used network protocols (HTTPS, TCP, UDP) to implement new features at central servers with association of database NoSQL.

- **Code Optimization & Congestion Management**

Handled a vast number of requests of more than 100 million users per day by permanently testing and optimizing the system. Managed troubleshoot server congestion in rush hours.

2018–2020 **Research Member**, ML4U Research Group, HCMUT, Vietnam

- **Experience sharing**

Participated in weekly seminars about Machine Learning coordinated by lecturers at HCMUT. Cooperated with a teammate to organize a seminar every two weeks for presentation

## Publications

- Do Duc Anh Nguyen, Pierre Alain, Fabien Autrel, Ahmed Bouabdallah, Jérôme François, and Guillaume Doyen. “How Fast Does Malware Leveraging EternalBlue Propagate? The case of WannaCry and NotPetya” SecSoft Workshop, 2024
- Do Duc Anh Nguyen, Pierre Alain, Fabien Autrel, Ahmed Bouabdallah, and Jérôme François. “Intent-based Attack Mitigation Through Opportunistic Synchronization of Microservices” NetSoft PhD Symposium, 2024
- Do Duc Anh Nguyen, Fabien Autrel, Ahmed Bouabdallah, and Guillaume Doyen. “A Robust Approach for the Detection and Prevention of Conflicts in I2NSF Security Policies.” NOMS 2023-2023 IEEE/IFIP Network Operations and Management Symposium. IEEE, 2023.

## Achivements & Scholarships

- **Full Studying Overseas Scholarship**, Vingroup, 2020.
- **Scholarships** for Outstanding Student, HCMUT, 2019.
- **Scholarships** for Talent Students in Honors Program, HCMUT 2016, 2017, 2018.
- **Highest Score Students** (Top 0.1%/300.000) in National High School Graduation Examination, 2015.

## Skills

- Programming Languages: Java, C/C++, Python, Ocaml.
- Languages: English (advanced), French (intermediate).
- Technology experience: MirageOS, OpenStack, GNS3, OMNet++

## Projects

- **Opportunistic Synchronization Mechanism**, 2025  
Implement a fast reaction mechanism that leverage existing traffic to deliver inforamtion between microservices which are deployed as security functions enable flexibility and scalability
- **Malware Propagation Testbed**, 2024  
Build a testbed to measure the propagation speed of WannaCry and NotPetya, and analyse their behavior
- **Policy Conflict Detection and Resolution for I2NSF framework**, 2023  
Implement a new conflict policy detection and resolution mechanism for the I2NSF framework proposed by the IETF. The mechanism operate at user level to provide meaningful feedback and suggestion for conflict resolution

## Social activities

- **Treasurer in ADER (Association de Doctorants de l'ENST de Bretagne, campus de Rennes):** Managed budget and participated in doctoral after work, Rennes (2023-2025)
- **Green Summer Voluntary Campaign:** Participated in building roads in a disadvantaged village in Ben Tre Province (2017)
- **Mid-autumn Festival:** Organized a Moon Festival for an elementary school with food and handmade lantern in Ben Tre province (2016, 2017)
- **The House of Love::** Contributed food, handmade toys to orphans and helped them to clean their home in Binh Duong province. (2015)

## References

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| ○ <b>Prof. Guillaume Doyen</b><br>SOTERN - IRISA<br>IMT Atlantique<br>guillaume.doyen@imt-atlantique.fr        | ○ <b>Prof. Christophe CLAVIER</b><br>Cryptis Program<br>University of Limoges, XLIM<br>christophe.clavier@unilim.fr  |
| ○ <b>Dr. Jérôme François</b><br>SnT<br>Univ. of Luxembourg and Inria Nancy Grand Est<br>jerome.francois@uni.lu | ○ <b>Assoc. Prof. Emmanuel Conchon</b><br>Cryptis Program<br>University of Limoges, XLIM<br>emmanuel.conchon@xlim.fr |